

Freight Rate Prediction — Machine Learning Assessment

The assessment asks for a model trained on the labeled development data, predictions for all 12,000 November validation loads, a fixed December 2025 prediction chart, a repository with code/dependencies/run instructions, and a short walkthrough.

1. Data exploration and quality

The development set has 48,000 rows covering 2025-01-01 to 2025-10-31. The target is `posted_rate`. Missing values were found in `weight` (300 rows) and `market_index` (374 rows). `Weight` is median-imputed and a missingness indicator is retained. The final model excludes `market_index` and `quote_signal` because those fields are unavailable in the fixed December chart inputs. `Distance` has a strong linear correlation with `posted_rate` ($r \approx 0.909$), so `distance` is a primary predictive feature, supplemented by `route`, `equipment`, `weight`, `coordinates`, and calendar effects.

2. Validation strategy

The split is chronological: January–September 2025 is used for model development and October 2025 is a holdout. This mirrors the actual forecasting setup more closely than a random split because November is future data. It also avoids mixing later pricing behavior into earlier training observations.

Metric	October holdout
Rows	4,853
MAE	\$107.99
RMSE	\$646.46
R ²	0.8211
Best iteration	289

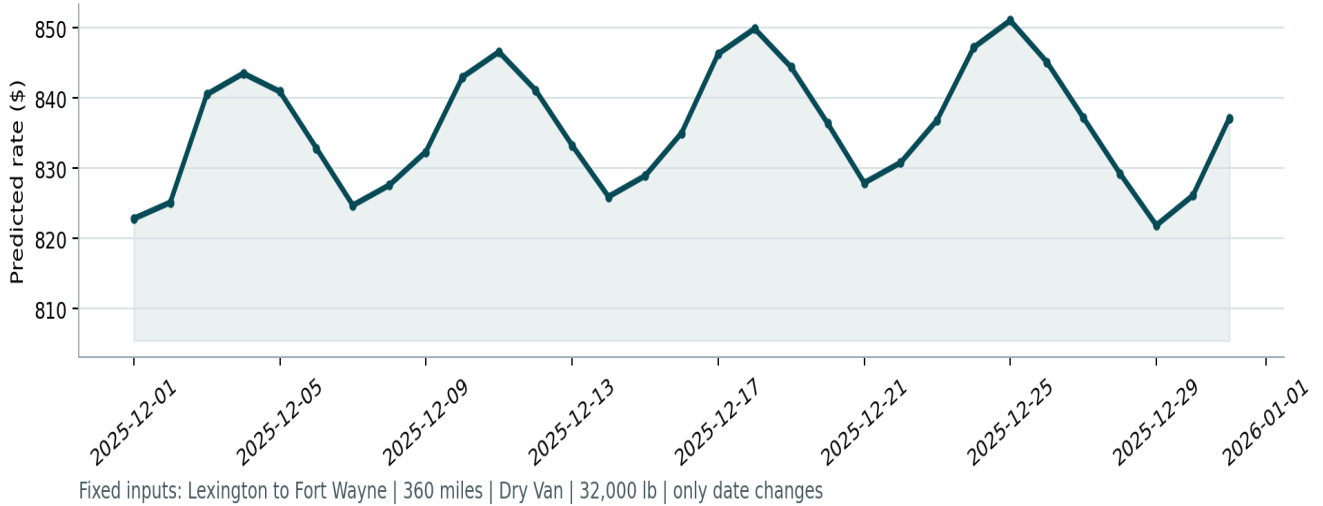
3. Feature engineering and model

CatBoostRegressor with MAE loss was selected because it handles categorical variables natively and can learn nonlinear interactions. Categorical features are `pickup`, `delivery`, `equipment`, and a combined `route`. Additional features include `log distance`, `weight per mile`, `distance-weight interaction`, `coordinate separation`, `date components`, and cyclical month/day-of-week features. The October holdout achieved MAE \$107.99, RMSE \$646.46, and R² 0.8211.

4. Final outputs and December chart

After validation, the model was refit on all 48,000 labeled development rows and used to predict all 12,000 validation loads. The December file contains the original seven required columns and one positive prediction for each day from December 1 through December 31, 2025. The supplied scorer validated both output files and generated the chart below.

Candidate: December 2025 Predicted Load Rate



5. Reproducibility

The package contains the training script, dependencies, README, scorer, model artifact, predictions, December inputs, and this report. The raw assessment data can remain local and need not be committed to GitHub.